$CANON TESTNET KEYS & $0 COST GUIDE

Full list of API keys, free public RPCs, worker keys & usage limits for Elysium Testnet (Chain 99801)

Prepared for Founders, Crowdfunders & Grant Reviewers | Zero Paid Subscriptions Needed

EST. LIVE BILL
$0.00 / MO

TOTAL MONTHLY BILL	EXTERNAL SIGNUPS	ZERO-KEY PUBLIC URLS	1-TAP LOCAL KEYS
$0.00 / mo	**2 Free IDs**	**5 Endpoints**	**4 Hex Keys**
100% permanent free tiers	Reown + Cloudflare Free	RPCs, Faucet & HL Info	Made free inside app

01. Free Account Signups (Only 2 Free Accounts Required)

SUBTOTAL: $0.00 / MO

VITE_WC_PROJECT_ID — Reown Cloud (WalletConnect) — $0.00 / mo

`Signup (Free): https://dashboard.reown.com -> Create Project -> Copy Project ID`

Pairs mobile & desktop wallets (MetaMask, Rabby, Rainbow, Trust) via QR or phone deep link. CANON already includes a working preview ID out of the box; grab your own free ID when hosting on your custom domain.

Free Limit: 1,000,000 relay msgs/mo | Est. Testnet Usage: ~3,000 - 12,000 msgs/mo (<1.2% used)

CLOUDFLARE + KV IDS — Cloudflare Workers & KV (Free Plan) — $0.00 / mo

`Signup (Free): https://dash.cloudflare.com/sign-up (No server or credit card needed)`

Hosts two zero-server workers: canon-passport-worker (checks Hyperliquid history & sponsors gasless HYPE Pass mints) and canon-keeper-worker (runs every 5 min to lock swap fees & buy the top 10 clean coins). Create PASSPORT_KV and KEEPER_KV via npx wrangler kv namespace create.

Free Limit: 100k req/day + 100k KV reads/day + 5 Crons | Est. Usage: ~400 - 800 req/day (<1% used)

GITHUB ACTIONS CRON — Optional Backup to Cloudflare Keeper — $0.00 / mo

`Included Workflow: .github/workflows/canon-keeper-cron.yml (https://github.com)`

Optional backup if you prefer GitHub over Cloudflare for the 5-minute keeper loop. Add KEEPER_KEY and CANON_DEPLOYMENT_JSON in GitHub Repo Settings -> Actions Variables & Secrets.

Free Limit: Unlimited minutes on public repos (2,000 min/mo private) | Est. Cost: $0.00

GOOD TO KNOW: ZERO PAID AI OR RPC KEYS NEEDED

You will see GEMINI_API_KEY listed as a blank template line inside .env.example from the studio host. Leave it blank - CANON does not call any paid AI APIs. All coin checks (Lens), basket math, swaps, staking, and votes run directly on free public chain data.

02. Public Blockchain & Data Endpoints (Zero Signup, No Key)

SUBTOTAL: $0.00 / MO

VITE_RPC_URL / ELYSIUM_RPC `https://testnet-rpc.elysium.kinetiq.xyz`
Official Elysium Testnet (Chain 99801) JSON-RPC for deploying contracts, trading, staking & ballots.
`Public free RPC | Est. 5k - 25k calls/mo | $0.00/mo`

Elysium Test HYPE Faucet `https://elysium.kinetiq.xyz/testnet-faucet`
Free Test HYPE gas drip. ~0.02 Test HYPE deploys all 8 contracts; ~0.10 Test HYPE funds 5,000+ worker runs.
`Official daily free drip | $0.00`

Ascend Test USDC Mint `0x4b4fb5C5794F0c821464956374cC5872D5F0f140`
Open mint() token on Elysium 99801 called automatically during Guided Seed to mint 500 Test USDC.
`Unlimited free testnet mints | $0.00`

HYPEREVM_RPC `https://rpc.hyperliquid.xyz/evm`
Public HyperEVM read RPC used by canon-passport-worker to read HYPE, kHYPE, WHYPE & KNTQ balances.
`Public read-only RPC | Cached 7 days in KV | $0.00/mo`

HL_INFO_URL `https://api.hyperliquid.xyz/info`
Public Hyperliquid L1 info endpoint used to score real user trading & staking history for 1-person-1-vote.
`Public REST endpoint | Zero API key required | $0.00/mo`

03. Local Wallet Keys (Made Free in 1 Tap Inside Launch $CANON)

SUBTOTAL: $0.00 / MO

Open Launch $CANON -> Full Controls -> Workers -> Create Fresh Worker Keys to make all 4 keys offline for free:

ATTESTOR_KEY Target: canon-passport-worker $0.00
32-byte hex key that signs verified HYPE Pass tier approvals. Uses 0 gas. Cost: $0.00.

RELAYER_KEY Target: canon-passport-worker $0.00
Pays free Test HYPE gas so users mint HYPE Passes for free. Fund once with ~0.05 Test HYPE. Cost: $0.00.

PEPPER_SALT Target: canon-passport-worker $0.00
Random 32-byte hex salt so users' main Hyperliquid wallets stay private on-chain. Cost: $0.00.

KEEPER_KEY Target: canon-keeper-worker $0.00
Runs pnthRouter.run() & indexVault.rebalance() every 5 min. Fund once with ~0.05 Test HYPE. Cost: $0.00.

TRIGGER_TOKEN (Opt.) Target: canon-keeper-worker $0.00
Optional random password you pick if triggering POST /run manually over HTTP. Cost: $0.00.

DEPLOYER_KEY (Opt.) Target: CLI deploy only $0.00
NOT needed for phone wallet Guided Launch. Only used if running npm run testnet:deploy on PC. Cost: $0.00.

04. Where Every Variable Goes (Quick Wiring Table)

ALL FREE ($0.00)

1. Web App & Server (.env) $0.00 / mo

`Vars: VITE_WC_PROJECT_ID, VITE_RPC_URL, PASSPORT_API, APP_URL`

Set VITE_WC_PROJECT_ID (Reown ID) + PASSPORT_API (your deployed canon-passport-worker *.workers.dev URL).

2. workers/passport-worker $0.00 / mo

`Vars: REGISTRY_ADDRESS, PASSPORT_KV, ATTESTOR_KEY, RELAYER_KEY, PEPPER_SALT`

Paste REGISTRY_ADDRESS + KV id in wrangler.toml; store the 3 hex values with npx wrangler secret put.

3. workers/keeper-worker $0.00 / mo

`Vars: CANON_DEPLOYMENT_JSON, KEEPER_KV, KEEPER_KEY`

Paste exported CANON_DEPLOYMENT_JSON + KV id in wrangler.toml; store KEEPER_KEY with npx wrangler secret put.

05. Active $CANON Elysium 99801 Contracts (Auto-Stamped From App)

CHAIN ID: 99801

$CANON Index Token (pnth)	`0x048aCeb334cb1835dc897a749C5985e169EAb1C5`
Fee Lock Router (pnthRouter)	`0xE0Fb49DBDC3406cF1f175EeCABDFb282a7a98b66`
10-Coin Index Vault (indexVault)	`0xE636003Feb05D0FB8f6016733C3d723446C9f221`
Staking Rewards (staking)	`0x30c88fce1fAD545Df0fC603287f93F0f94076224`
Basket Registry (constituentRegistry)	`0xD8cEA0015E2C876ab5fa6D9178b46B57d65c6087`
HYPE Pass Registry (passportRegistry)	`0x0Eb4A8eB47364Be9d18410AF2f850db92b807290`
250 $CANON Backer Drip Faucet (drip)	`Created in Guided Step 3`

06. Simple 4-Step Go-Live Order (Mobile First)

TOTAL: $0.00

01. Open Launch $CANON -> Guided Launch on your phone, claim free Test HYPE gas, and deploy + seed $CANON ($0).

02. Create a free Reown Project ID at dashboard.reown.com when moving to your own domain ($0).

03. In Full Controls -> Workers, tap Create Fresh Worker Keys, fund Relayer & Keeper with 0.05 Test HYPE, and deploy to Cloudflare Free ($0).

04. Set PASSPORT_API to your Worker URL and tap Copy Live Backer Link to share with crowdfunders & grant teams ($0).